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course/program:- BTech CSE SY

Subject:- DLM

DLM assignment 1.

Q.1) Subtract $(1011)_2$ from $(1110)_2$ using the 1st complement method.

Solⁿ:- $1110 \rightarrow 14$
 $1011 \rightarrow 11$ $14 - 11 = 3$

$11 \rightarrow 1011 \rightarrow 0100$ (1st complement).

$$\begin{array}{r} 1110 \\ + 0100 \\ \hline 10010 \end{array}$$

To get positive Answer

$$\begin{array}{r} 0010 \\ + 1 \\ \hline 0011 \end{array}$$

$0011 \rightarrow 3$

Q.2) Subtract $(1110)_2$ from $(1011)_2$ using the 1st complement method.

Solⁿ:- $1011 \rightarrow 11$
 $1110 \rightarrow 14$ $11 - 14 = -3$

$14 \rightarrow 1110 \rightarrow 0001$

$$\begin{array}{r} + 1011 \\ 0001 \\ \hline 1100 \end{array}$$

$1100 = -3$



Q.3) $(16)_{16} = (?)_{10}$

Solⁿ:- Hexadecimal to decimal.

$$1 \times 16^1 + 6 \times 16^0 = 16 + 6 \Rightarrow 22$$

$$(16)_{16} \rightarrow (22)_{10}$$

Q.4) $(187)_{10} = (?)_{12}$

2	187	1
2	93	1
2	46	0
2	23	1
2	11	1
2	5	1
2	2	1
2	2	0
	1	

$$(10111011)_2$$

Q.5) $(11111110010)_2 = (?)$

Binary to Hexadecimal

<u>1111</u>	<u>1111</u>	<u>0010</u>	$= (FF2)_{16}$
↓	↓	↓	
F	F	2	



Q.6) $(01011)_2 = (?)$

Binary to Decimal

$$0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ \Rightarrow (11)_{10}$$

Q.7) $(1001.0010)_2 = (?)_{10}$

Binary to Decimal

$$1 \times 2^3 + 1 \times 2^0 = 9 \quad \text{--- ①}$$

$$1 \times 2^{-3} = 0.125 \quad \text{--- ②}$$

$$\text{Add ① and ②} \Rightarrow 9.125$$

Q.8) The correct BCD representation of the Excess 3 no.
10107.

Solⁿ:-

Step:1) Arrange the Excess 3 digit.

Step:2) make 4 pairs of each digit.

Step:3) convert it into decimal and subtract 3 from it

Step:4) After subtraction the final answer is the BCD

$$\begin{array}{r} 1010 \\ \hline \downarrow \end{array}$$

$$10$$

$$\begin{array}{r} 10 \\ - 3 \\ \hline 7 \end{array}$$

7 is the BCD



Q.9) what is the correct Excess representation of the BCD number 1101?

Solⁿ:-

- Step:1) Arrange the BCD digit
Step:2) make 4 digit as a set
Step:3) convert it into decimal form
Step:4) Add 3 into it and then convert it into binary form

$$\begin{array}{c} 1101 \\ \hline \downarrow \end{array}$$

16 is valid

$$\begin{array}{r} 1 \\ + 3 \\ \hline 4 \end{array}$$



0100

$$\begin{array}{r} 3 \\ + 3 \\ \hline 6 \end{array}$$

0110

Q.10) The gray code equivalent of the decimal number 5 is 7.

Solⁿ:-

- Step:1) Record the MSB as it is
Step:2) Add the MSB to the next bit
Step:3) Repeat the steps

$$5 \rightarrow 0101 \rightarrow \begin{array}{r} 0101 \\ \downarrow \downarrow \downarrow \\ 010 \\ \hline 0111 \end{array}$$

$\Rightarrow (0111)_2$

Q.11) what is the gray code equal to 14?

Solⁿ:-

- Step:1) Record the MSB as it is
Step:2) Add the MSB to the next bit
Step:3) Repeat the steps

$$14 \rightarrow 1110$$

$$\begin{array}{r} 1110 \\ \downarrow \downarrow \downarrow \\ 1001 \end{array}$$

$\Rightarrow (1001)_2$

Full marks